



TECHNICAL DATA SHEET

ProLine ASA-CF Filament

Product Overview

ProLine ASA-CF is an ASA-based 3D printing filament with carbon-fiber reinforcement. It is designed for functional parts that require good stiffness, dimensional stability and weather resistance, making it suitable for applications where UV and outdoor exposure matter.

Key Features

- Rigid, dimensionally stable parts
- Strong, durable components
- Performs reliably at elevated temperatures
- Weather resistant for outdoor use

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm³	1.09–1.10
MFR (250°C / 2.16 kg)	ISO 1133	g/10 min	4–7
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.5
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa	39–42
Elongation at break (X-Y)	ISO 527	%	15.5–17.5
Flexural modulus (X-Y)	ISO 527	MPa	850–900
Flexural strength (X-Y)	ISO 178	MPa	80–82.5
Impact strength (X-Y)	ISO 180	kJ/m²	10.5–12
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	88
Continuous Use Temperature	IEC 60216	°C	85

Recommended Printing Parameters

Parameter	Setting
Print Temperature	290-310°C
Bed Temperature	100-120°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	0-20%
Printing Environmental Temperature	Room temperature to ~60°C

Drying & Handling Notes

- **Pre-dry: 80°C, minimum 5 h (dryer) to ensure print success rate and model quality**
- **Post-dry: 80-100°C, 1-3 h to increase part strength**

Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

Test Specimens

Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

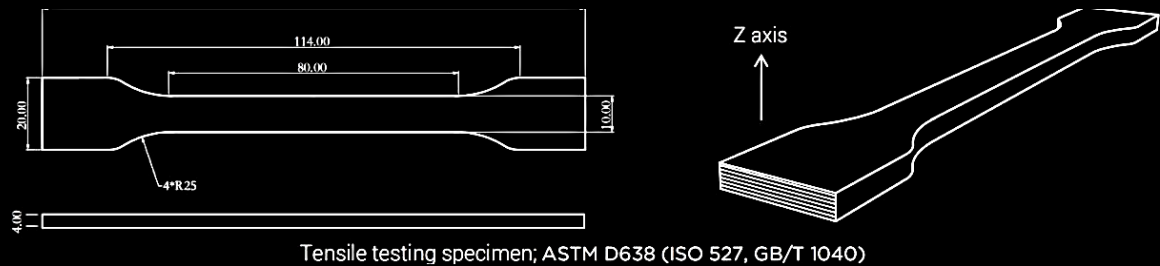


Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)

